MY EXPERIENCES WITH .PER FILES

By Phill Phree.

Introduction

OK, let's start simple. There may be some who don't know this, so if you do don't feel patronised. I make no assumptions about people's knowledge in this article and hope there will be something here that all can find useful.

Two files are used to control a Computer Player (CP), the AI and the PER. Despite its name, the AI is **not** the artificial intelligence file and does **not** control how the computer player behaves. The AI file simply contains the list of buildings, units and technologies you wish the CP to make or research, and will be dealt with elsewhere. How the CP uses the things it has made and researched, and how it behaves towards other players, is governed by the PER (Personality) file.

Before I continue I must state the following – THESE ARE GUIDELINES ONLY. They represent my experience from making campaigns for over a year now, and 3 months study on the files themselves, so I hope they are informed, but many aspects of PERs remain a mystery as parts of them have never been fully documented.

Also, it is not my intention to display the meanings of all the numbers in the PER file here, that has been done before – note, however, that the Ensemble Studios / Microsoft documentation is vague, incomplete and almost utterly worthless as a reference. The best reference I have found is PerEdit by Martin Sturm, which can be found at Age Of Empires Heaven, on the Granary in the Utilities section. It contains descriptions for the numbers (as far as they are known) and allows you to create your own PERs by selecting the traits you want from drop-down lists, but I find it quicker and easier to create the PER files in Notepad and just use PerEdit as a reference. The purpose of this document is to try and go one step further and attempt to help you create PERs that are customised to your needs, by pointing out some of the personality traits that I have found particularly useful (and not so useful).

So, with that said, let's get down to it.

Getting started

The PER file contains all the personality traits you wish the CP to display, and does so by the use of Strategic Numbers (SNs). In Appendix 1 are the numbers themselves so you can copy and paste the traits you want into a new file for your personal use, and in Appendix 2 are the acceptable values for each SN.

Each SN is a number pair i.e. the first number represents the trait, the second represents how strongly or otherwise that trait is implemented. An example is below (// denotes a comment for clarity – should you create and edit your own PERs you must put // in front of any comments, failure to do so may crash the game);

```
170     500     //SNMinimumFood
```

This means the computer will try and keep 500 food on hand at all times. The 500 can be changed to anything you wish in this case.

Other SNs accept a range of values, for instance;

49 1 //SNRetreatAfterTargetDestroyed

This can take 4 values – 1 means retreat if no other targets are in range, 2 means always retreat and 3 means seek out and destroy new targets (or extermination mode as it is called). 0 means never retreat and regroup at the current point.

Others simply accept a value of 1 to turn them on or 0 to disable them, e.g.

196 //SNCoopShareAttacking

Set to 1, this allows CPs to attack to defend each other. Set to 0 it's every CP for himself ;-)

There are also numbers that depend on other numbers being implemented first, e.g.

103 //SNAttackIntelligence

This is 1 for on and 0 for off, and decides whether or not to use the intelligent attack system, whatever that means. It is a vague description that really says very little, and is typical of the documentation for these files. But we do know that it can be used with another SN, as follows;

103 1 //SNAttackIntelligence
47 //SNAttackCoordination

With the Attack Intelligence 'switched on', the Attack Coordination makes the CP smarter still when attacking you. If SN47 is set to 0 (or the SN is simply omitted from the PER file) then the attacks are not coordinated, but should still be relatively intelligent (for a CP). Set to 1, it allows the CP to attack with 1 coordinated group. Set to 2, it allows the CP to attack with multiple groups, and can therefore allow the CP to use multi-front attacks. This is the option I use pretty much all the time.

Of course you must make sure you have told the CP to have enough attack groups to do this with i.e. at least 2. Another point I would make here is to ensure, in your AI file, that you are making enough units to cover this, i.e. if you want 3 groups of at least 4 soldiers you must make a minimum of 12 in the AI. I know this seems obvious but in my experience the obvious is often overlooked.

Some Useful Numbers

I could easily turn this section into a dissertation so I'll just give a few examples here. Hopefully it will give you some idea of the things you can do to help guide your CP along the route to becoming an efficient and clever enemy, within the constraints of the game.

<u>How aggressive?</u>

You may want the CP to keep attacking with no regard for its soldiers, perhaps because you have given the CP plenty of resources and that's the nature of the enemy. Or you may want it to hit and pull back, to simulate raiding. The following numbers help to decide how aggressive or defensive a CP can be;

30 //SNPercentHealthRetreat

31 //SNPercentDeathRetreat
91 //SNPercentUnitHealthRetreat

The lower the values the more defensive. Higher values make good aggressive troops, and make great suicide squads. However, just because they are aggressive doesn't mean they have to be stupid. They can still be pretty smart if they retreat after destroying targets, have attack intelligence, attack on more than one front etc. And consider numbers such as;

75 2 //SNGroupCommanderSelectionMethod
121 1 //SNDefendImportantGroupLeaders

I set these options for a CP who was Egyptian, resulting in a group of Scythe Chariots with Priests at the back. A tough group to beat, especially with the awesome range those guys have.

The Autobuild Set

Sometimes you don't need to specifically tell the AI file to build. The Autobuild numbers in the PER are all very simple to use, they are simply 1 or 0. Here they are;

180 //SNAutoBuildHouses
205 //SNAutoBuildDropsites
206 //SNAutoBuildFarms
207 //SNAutoBuildTowers
208 //SNAutoBuildDocks
209 //SNAutoBuildFishingBoats
210 //SNAutoBuildTransports
223 //SNAutoBuildWarships

I never got the last one to work, so I always make warships in the AI instead. But the others have all worked well for me. This brings up an issue in itself. I remember Ingo saying once that he hated the way the CP made houses and spoilt his maps. When maps are made by a genius such as Ingo, the frustration is not hard to see. Simply give 180 a value of 0 to stop it happening. There are times when what you deliberately restrict the CP from doing is just as important as what you specifically tell it to do.

Tribute (and Diplomacy)

Lots of designers like playing with the tribute numbers because it's the only way to get the CP to 'speak'. But they have other uses too. For example, in a scenario I designed I did not want the player to tribute the CP because the story relies on them being destroyed, so to ensure the gameplay did not conflict with the campaign story I simply did the following;

217 0 //SNAllowDiplomacyChangeOnTribute

So no matter how much you give the CP it will still consider you an enemy. This is a diplomacy number, not a tribute number, but is very handy. Of course setting it to 1 can create interesting situations when there is more than one CP in the scenario, allowing you to possibly ally with one CP against another and get some help if you are having a hard time – I say possibly because the other CP(s) may try and tribute them too. For the record, when CPs tribute each other they often tribute a CP with a higher technology score, so alliances can be guessed sometimes from the achievements screen.

Another very useful number is;

215 0 //SNAllowDiplomacyChangeOnAllyAttack

In my last campaign, in the last scenario, the player had a CP ally and fought against another CP. Just in case you hit your allies with e.g. a catapult stone by accident, it stopped them turning on you.

Getting resources

86 //SNStoragePitMaxDistance
87 //SNGranaryMaxDistance

These values are very useful if you don't want your CP making drop sites miles away from the resource they are storing.

204 //SNMinimumElephantHuntGroupSize

In RoR the CP villagers don't seem inclined to hunt elephants anyway, but if they do there is no point in them being suicidal, so make this around 4.

4 //SNCapCivilianBuilders
5 //SNCapCivilianGatherers

Sometimes the CP may be keener on resource gathering or building than attacking, so use these numbers to define the maximum number of villagers it can use for these tasks to stop it building endless peons.

Problems

Sometimes no matter what you do the CP just sits there with its army and won't attack unless you go to them first. Other times you tell it to be defensive and it throws troops at you like terminally depressed lemmings. Consider the following SN;

20 //SNEnemySightedResponseDistance

This value should decide the range within which the CP will attack once it has spotted you. However, the maximum value, according to the documentation, is 144 tiles. When RoR came out it introduced Gigantic Maps but the PER values were never updated to accommodate this, so if you are at one end of the map and the CP is at the other, they may just sit there and do nothing. Ironically I have had them attack when they are on an island and so am I, on a Gigantic Map at opposite ends, and they have dropped transports full of troops on me, so you can never tell. But it may explain why sometimes they are static. Or not, who knows?

Another problem is that, even on high path finding and hardest level, both of which bring out the best in PERs, if you have created a scenario with limited routes to the enemy the CP may not be very efficient with its attacks. Troops getting stuck in trees, or not being able to find you, or trying to send a big group through a small area and the CP 'giving up', all these have happened to me in scenarios I have designed and are incredibly frustrating. The only thing you can do in these cases is change the map a little to provide slightly better access and possibly play around with the SNs that govern exploring. Also, changes to a map can dramatically change the CP behaviour, so be aware of that, even placing or deleting a few trees can make a difference depending on the situation.

The bottom line here is TEST, TEST, TEST. My philosophy on this is that if I am not sick of the sight of it then I have not tested it enough. If you can find people willing to help you playtest (not usually a problem) so much the better; despite the fact that we all run and play the same game, effects on other peoples' machines sometimes differ from your own. I had a scenario where the CP sat there and did nothing, only to be told by one of my playtesters that he found them very aggressive and quick to attack. I have no idea why, maybe it has something to do with the patches people have installed for the game but having others try your stuff may save you from unnecessary time and effort trying to fix something that isn't broken.

There is a very useful utility, the Speed Editor, again from the Granary at AoEH, which allows you to speed up the whole game, and it works with both AoE and RoR. The highest speed you can set in the game itself is 'very fast', which is a speed of 20 in the Speed Editor. Run the Speed Editor before running the game, you must be out of the game to change the speed. Usually when testing I set it to about 50, this allows me to quickly see the effects of small adjustments to PER files without having to wait for hours, and, with PerEdit, gives me the best possible set of tools for working with these files. Why spend days testing your stuff when you can do it in hours? Be aware though that if you set the speed too high the game will be unplayable and what you see will not necessarily be representative of what you have changed. A speed of 40-60 is about right, depending on the size of the map. It is also very useful for checking that your AI files are working correctly i.e. building what you want them to, in the right order etc. Playtest your scenarios with 'reveal map' and 'no fog' with a speed of 40-60 and you will quickly be able to monitor and adjust the PER and AI files as necessary.

And good luck. Anyone who has good results, bad results, questions, corrections or additions, please feel free to mail me at pphree@patdist.freeserve.co.uk

Appendix 1 – The SNs.

N.B. These are the 'bare' numbers, putting in values for them would be 'defining a personality' but they are here because some, like me, may prefer to copy and paste the ones they need and add their own values to them, rather than selecting them from lists in PerEdit, a process I find slower and more tedious than doing it myself. No offence is meant to Martin Sturm by that comment, I am very grateful to him for his excellent program, it's just my personal preference.

All PERs must start with DEFAULT and end with END.

```
DEFAULT
//CIVILIAN NUMBERS
0       //SNPercentCivilianExplorers
1       //SNPercentCivilianBuilders
2       //SNPercentCivilianGatherers
3       //SNCapCivilianExplorers
4       //SNCapCivilianBuilders
5       //SNCapCivilianGatherers
18      //SNTotalNumberExplorers
35      //SNMinimumCivilianExplorers
117     //SNFoodGathererPercentage
118     //SNGoldGathererPercentage
119     //SNStoneGathererPercentage
120     //SNWoodGathererPercentage
147     //SNMostNeededResourceLookAhead
145     //SNNumberEnemyObjectsRequired
148     //SNRetaskGatherAmount
149     //SNMaxRetaskGatherAmount
167     //SNInitialExplorationRequired
203     //SNUseByTypeMaxGathering
204     //SNMinimumElephantHuntGroupSize

//GROUP-RELATED NUMBERS
40      //SNGroupFillMethod
75      //SNGroupCommanderSelectionMethod
76      //SNConsecutiveIdleUnitLimit
143     //SNTaskUngroupedSoldiers

//ATTACK GROUP NUMBERS
36      //SNNumberAttackGroups
37      //SNAttackGroupMakeup
16      //SNMinimumAttackGroupSize
26      //SNMaximumAttackGroupSize
93      //SNScaleMinimumAttackGroupSize
94      //SNScaleMaximumAttackGroupSize
98      //SNAttackGroupSizeRandomness
41      //SNAttackGroupGatherSpacing
30      //SNPercentHealthRetreat
31      //SNPercentDeathRetreat
91      //SNPercentUnitHealthRetreat
95      //SNScalePercentHealthRetreat
96      //SNScalePercentDeathRetreat
```

```
97      //SNScalePercentUnitHealthRetreat
121     //SNDefendImportantGroupLeaders
131     //SNGroupLeaderDefenseDistance

//MISCELLANEOUS ATTACK NUMBERS
46      //SNAttackSeparationTime
47      //SNAttackCoordination
19      //SNPercentEnemySightedResponse
20      //SNEnemySightedResponseDistance
48      //SNAttackResponseSeparationTime
49      //SNRetreatAfterTargetDestroyed
135     //SNBlotExplorationMap
136     //SNBlotSize
71      //SNLockAttackAndAttackResponse
100     //SNMaximumGAIAAttackResponse
102     //SNAttackSeparationTimeRandomness
103     //SNAttackIntelligence
104     //SNInitialAttackDelay
134     //SNInitialAttackDelayType

//DEFEND GROUP NUMBERS
38      //SNNumberDefendGroups
39      //SNDefendGroupMakeup
25      //SNMinimumDefendGroupSize
28      //SNMaximumDefendGroupSize
50      //SNGoldDefendPriority
51      //SNStoneDefendPriority
52      //SNForageDefendPriority
53      //SNChokePointDefendPriority
54      //SNRuinsDefendPriority
55      //SNArtifactDefendPriority
56      //SNTownDefendPriority
57      //SNDefenseDistance
22      //SNSentryDistance
72      //SNSentryDistanceVariation
92      //SNDefendOverlapDistance

//EXPLORE GROUP NUMBERS
42      //SNNumberExploreGroups
43      //SNMinimumExploreGroupSize
44      //SNMaximumExploreGroupSize
45      //SNExploreGroupMakeup

//BOAT ATTACK GROUP NUMBERS
58      //SNNumberBoatAttackGroups
59      //SNMinimumBoatAttackGroupSize
60      //SNMaximumBoatAttackGroupSize

//BOAT DEFEND GROUP NUMBERS
67      //SNNumberBoatDefendGroups
68      //SNMinimumBoatDefendGroupSize
69      //SNMaximumBoatDefendGroupSize
```

70 //SNDockDefendPriority

//BOAT EXPLORE GROUP NUMBERS
61 //SNNumberBoatExploreGroups
62 //SNMinimumBoatExploreGroupSize
63 //SNMaximumBoatExploreGroupSize
64 //SNDesiredNumberTradeEscorts
65 //SNDesiredNumberFishEscorts
66 //SNDesiredNumberTransportEscorts

//TOWN BUILDING NUMBERS
73 //SNMinimumTownSize
74 //SNMaximumTownSize
86 //SNStoragePitMaxDistance
87 //SNGranaryMaxDistance
84 //SNNumberWallGates
85 //SNSizeWallGates
112 //SNMinimumWaterBodySizeForDock
180 //SNAutoBuildHouses
205 //SNAutoBuildDropsites
206 //SNAutoBuildFarms
207 //SNAutoBuildTowers
208 //SNAutoBuildDocks
209 //SNAutoBuildFishingBoats
210 //SNAutoBuildTransports
223 //SNAutoBuildWarships
212 //SNDesiredNumberDocks
213 //SNDesiredNumberFishingBoats
214 //SNDesiredNumberTransports
224 //SNDesiredNumberWarships
174 //SNMaximumHousesBeforeDropsites
175 //SNSpecificBuildItemToBuild
176 //SNSpecificBuildItemTime
177 //SNUnskippableItemType
150 //SNMaxStoragePits
151 //SNMaxGranaries
152 //SNHouseOverage
155 //SNBuildPlanDivisions
160 //SNMaxBuildPlanGathererPercentage
161 //SNRequiredFirstBuilding
163 //SNFoodDropsiteDistance
164 //SNWoodDropsiteDistance
165 //SNStoneDropsiteDistance
166 //SNGoldDropsiteDistance
220 //SNMaxFarms
222 //SNMaxTowers
202 //SNMinimumDropsiteBuffer
168 //SNRandomPlacementFactor
169 //SNMinimumForestTiles
181 //SNUpgradeToToolAgeASAP
182 //SNUpgradeToBronzeAgeASAP
183 //SNUpgradeToIronAgeASAP

//TARGET EVALUATION NUMBERS
77 //SNTargetEvaluationDistance
78 //SNTargetEvaluationHitpoints
79 //SNTargetEvaluationDamageCapability
80 //SNTargetEvaluationKills
81 //SNTargetEvaluationAllyProximity
82 //SNTargetEvaluationROF
83 //SNTargetEvaluationRandomness
89 //SNTargetEvaluationAttackAttempts
90 //SNTargetEvaluationRange
106 //SNSpecialAttackType1
109 //SNSpecialAttackInfluence1
107 //SNSpecialAttackType2
110 //SNSpecialAttackInfluence2
108 //SNSpecialAttackType3
111 //SNSpecialAttackInfluence3
122 //SNTargetEvaluationContinent
123 //SNTargetEvaluationSiegeWeapon
144 //SNTargetEvaluationBoat
184 //SNTargetEvaluationTimeKillRatio
185 //SNTargetEvaluationInProgress

//COMPUTER PLAYER TRIBUTE NUMBERS
124 //SNTributeAmount
125 //SNTributeChatFrequency
126 //SNTributeChatRandomness
127 //SNTributeTimeout
128 //SNTributePlayer
129 //SNTributeSuccessOutcome
130 //SNTributeFailureOutcome
132 //SNTributePersistence
133 //SNTributeRevokeOnAttack

//MISCELLANEOUS NUMBERS
23 //SNArtifactReturnDistance
29 //SNMinimumPeaceLikeLevel
32 //SNPercentExplorationRequired
34 //SNZeroPriorityDistance
88 //SNTacticalUpdateFrequency
99 //SNScalingFrequency
101 //SNBuildFrequency
105 //SNSaveScenarioInformation
114 //SNNumberBuildAttemptsBeforeSkip
115 //SNMaxSkipsPerAttempt
170 //SNMinimumFood
171 //SNMinimumWood
172 //SNMinimumStone
173 //SNMinimumGold
190 //SNMaximumFood
191 //SNMaximumWood
192 //SNMaximumStone

```
193     //SNMaximumGold
215     //SNAllowDiplomacyChangeOnAllyAttack
216     //SNMinimumAmountForTrading
217     //SNAllowDiplomacyChangeOnTribute
221     //SNHitsBeforeAllianceChange
178     //SNAttackDiplomacyImpact
218     //SNEasiestReactionPercentage
219     //SNEasierReactionPercentage
201     //SNTrackPlayerHistory
188     //SNAttackWinningPlayer
194     //SNCoopShareInformation
195     //SNAttackWinningPlayerFactor
196     //SNCoopShareAttacking
197     //SNCoopShareAttackingInterval
198     //SNPercentageExploreExterminators
186     //SNCoopDemandTributeInterval
187     //SNCoopDemandTributeMaximum
END
```

Appendix 2 – SN List of accepted values

	DESCRIPTION	RANGE
//CIVILIAN NUMBERS		
0	//SNPercentCivilianExplorers	0-100
1	//SNPercentCivilianBuilders	0-100
2	//SNPercentCivilianGatherers	0-100
3	//SNCapCivilianExplorers	>=0
4	//SNCapCivilianBuilders	>=0
5	//SNCapCivilianGatherers	>=0
18	//SNTotalNumberExplorers	>=0
35	//SNMinimumCivilianExplorers	>=0
117	//SNFoodGathererPercentage	0-100
118	//SNGoldGathererPercentage	0-100
119	//SNStoneGathererPercentage	0-100
120	//SNWoodGathererPercentage	0-100
147	//SNMostNeededResourceLookAhead	>=0
145	//SNNumberEnemyObjectsRequired	>=0
148	//SNRetaskGatherAmount	>=0
149	//SNMaxRetaskGatherAmount	>=0
167	//SNInitialExplorationRequired	>=0
203	//SNUseByTypeMaxGathering	0,1
204	//SNMinimumElephantHuntGroupSize	>=0
//GROUP RELATED NUMBERS		
40	//SNGroupFillMethod	0,1
75	//SNGroupCommanderSelectionMethod	0,1,2

76	//SNConsecutiveIdleUnitLimit	>=0
143	//SNTaskUngroupedSoldiers	0,1

//ATTACK GROUP NUMBERS

36	//SNNumberAttackGroups	>=0
37	//SNAttackGroupMakeup	>=0
16	//SNMinimumAttackGroupSize	>=0
26	//SNMaximumAttackGroupSize	>=0
93	//SNScaleMinimumAttackGroupSize	>=0
94	//SNScaleMaximumAttackGroupSize	>=0
98	//SNAttackGroupSizeRandomness	>=0
41	//SNAttackGroupGatherSpacing	>=1
30	//SNPercentHealthRetreat	0-100
31	//SNPercentDeathRetreat	0-100
91	//SNPercentUnitHealthRetreat	0-100
95	//SNScalePercentHealthRetreat	>=0
96	//SNScalePercentDeathRetreat	>=0
97	//SNScalePercentUnitHealthRetreat	>=0
121	//SNDefendImportantGroupLeaders	0,1
131	//SNGroupLeaderDefenseDistance	>=0

//MISCELLANEOUS ATTACK NUMBERS

46	//SNAttackSeparationTime	>=0
47	//SNAttackCoordination	0,1,2
19	//SNPercentEnemySightedResponse	0-100
20	//SNEnemySightedResponseDistance	0-144
48	//SNAttackResponseSeparationTime	>=0
49	//SNRetreatAfterTargetDestroyed	0,1,2,3
135	//SNBlotExplorationMap	0,1
136	//SNBlotSize	0 -> Map Size
71	//SNLockAttackAndAttackResponse	0,1
100	//SNMaximumGAIAAttackResponse	>=0
102	//SNAttackSeparationTimeRandomness	>=0
103	//SNAttackIntelligence	0,1
104	//SNInitialAttackDelay	>=0
134	//SNInitialAttackDelayType	0,1,2,3

//DEFEND GROUP NUMBERS

38	//SNNumberDefendGroups	>=0
39	//SNDefendGroupMakeup	>=0
25	//SNMinimumDefendGroupSize	>=0
28	//SNMaximumDefendGroupSize	>=0
50	//SNGoldDefendPriority	0-7[i]

51	//SNStoneDefendPriority	0-7[(i)]
52	//SNForageDefendPriority	0-7[(i)]
53	//SNChokePointDefendPriority	0-7[(i)]
54	//SNRuinsDefendPriority	0-7[(i)]
55	//SNArtifactDefendPriority	0-7[(i)]
56	//SNTownDefendPriority	0-7[(i)]
57	//SNDefenseDistance	>=0
22	//SNSentryDistance	>=0
72	//SNSentryDistanceVariation	>=0
92	//SNDefendOverlapDistance	>=0

//EXPLORE GROUP NUMBERS

42	//SNNumberExploreGroups	>=0
43	//SNMinimumExploreGroupSize	>=0
44	//SNMaximumExploreGroupSize	>=0
45	//SNExploreGroupMakeup	>=0

//BOAT ATTACK GROUP NUMBERS

58	//SNNumberBoatAttackGroups	>=0
59	//SNMinimumBoatAttackGroupSize	>=0
60	//SNMaximumBoatAttackGroupSize	>=0

//BOAT DEFEND GROUP NUMBERS

67	//SNNumberBoatDefendGroups	>=0
68	//SNMinimumBoatDefendGroupSize	>=0
69	//SNMaximumBoatDefendGroupSize	>=0
70	//SNDockDefendPriority	0-7[(i)]

//BOAT EXPLORE GROUP NUMBERS

61	//SNNumberBoatExploreGroups	>=0
62	//SNMinimumBoatExploreGroupSize	>=0
63	//SNMaximumBoatExploreGroupSize	>=0
64	//SNDesiredNumberTradeEscorts	>=0
65	//SNDesiredNumberFishEscorts	>=0
66	//SNDesiredNumberTransportEscorts	>=0

//TOWN BUILDING NUMBERS

73	//SNMinimumTownSize	>=0
74	//SNMaximumTownSize	>=0
86	//SNStoragePitMaxDistance	>=0
87	//SNGranaryMaxDistance	>=0
84	//SNNumberWallGates	>=0
85	//SNSizeWallGates	>=0

112	//SNMinimumWaterBodySizeForDock	>=10
205	//SNAutoBuildDropsites	0,1
206	//SNAutoBuildFarms	0,1
207	//SNAutoBuildTowers	0,1
208	//SNAutoBuildDocks	0,1
209	//SNAutoBuildFishingBoats	0,1
210	//SNAutoBuildTransports	0,1
223	//SNAutoBuildWarships	0,1
212	//SNDesiredNumberDocks	>=0
213	//SNDesiredNumberFishingBoats	>=0
214	//SNDesiredNumberTransports	>=0
224	//SNDesiredNumberWarships	>=0
174	//SNMaximumHousesBeforeDropsites	>=0
175	//SNSpecificBuildItemToBuild	Item Id[ii]
176	//SNSpecificBuildItemTime	>=0
177	//SNUnskippableItemType	Item Id[ii]
150	//SNMaxStoragePits	>=0
151	//SNMaxGranaries	>=0
152	//SNHouseOverage	>=0
155	//SNBuildPlanDivisions	>=1
160	//SNMaxBuildPlanGathererPercentage	0-100
161	//SNRequiredFirstBuilding	Item Id[ii]
163	//SNFoodDropsiteDistance	>=3
164	//SNWoodDropsiteDistance	>=3
165	//SNStoneDropsiteDistance	>=3
166	//SNGoldDropsiteDistance	>=3
220	//SNMaxFarms	>=0
222	//SNMaxTowers	>=0
202	//SNMinimumDropsiteBuffer	0,1
168	//SNRandomPlacementFactor	>=0
169	//SNMinimumForestTiles	>=0
180	//SNAutoBuildHouses	0,1
181	//SNUpgradeToToolAgeASAP	0,1
182	//SNUpgradeToBronzeAgeASAP	0,1
183	//SNUpgradeToIronAgeASAP	0,1

//TARGET EVALUATION NUMBERS

77	//SNTargetEvaluationDistance	[iii]
78	//SNTargetEvaluationHitpoints	[iii]
79	//SNTargetEvaluationDamageCapability	[iii]
80	//SNTargetEvaluationKills	[iii]
81	//SNTargetEvaluationAllyProximity	[iii]
82	//SNTargetEvaluationROF	[iii]

83	//SNTargetEvaluationRandomness	(iii)
89	//SNTargetEvaluationAttackAttempts	(iii)
90	//SNTargetEvaluationRange	(iii)
106	//SNSpecialAttackType1	Item Id(ii)
109	//SNSpecialAttackInfluence1	>=0
107	//SNSpecialAttackType2	Item Id(ii)
110	//SNSpecialAttackInfluence2	>=0
108	//SNSpecialAttackType3	Item Id(ii)
111	//SNSpecialAttackInfluence3	>=0
122	//SNTargetEvaluationContinent	(iii)
123	//SNTargetEvaluationSiegeWeapon	(iii)
144	//SNTargetEvaluationBoat	(iii)
184	//SNTargetEvaluationTimeKillRatio	(iii)
185	//SNTargetEvaluationInProgress	>=0

//COMPUTER PLAYER TRIBUTE NUMBERS

124	//SNTributeAmount	>=0
125	//SNTributeChatFrequency	>=0
126	//SNTributeChatRandomness	0-<SNTributeChatFrequency
127	//SNTributeTimeout	>=0
128	//SNTributePlayer	Player No.
129	//SNTributeSuccessOutcome	0,1
130	//SNTributeFailureOutcome	0,1
132	//SNTributePersistence	0,1
133	//SNTributeRevokeOnAttack	0,1

//MISCELLANEOUS NUMBERS

23	//SNArtifactReturnDistance	>=0
24	//SNPercentVictoryClamp	0-100
29	//SNMinimumPeaceLikeLevel	0-100
32	//SNPercentExplorationRequired	0-100
34	//SNZeroPriorityDistance	1-144
88	//SNTacticalUpdateFrequency	>=0
99	//SNScalingFrequency	>=0
101	//SNBuildFrequency	>=0
105	//SNSaveScenarioInformation	0,1
114	//SNNumberBuildAttemptsBeforeSkip	>=1
115	//SNMaxSkipsPerAttempt	>=1
170	//SNMinimumFood	>=0
171	//SNMinimumWood	>=0
172	//SNMinimumStone	>=0
173	//SNMinimumGold	>=0
190	//SNMaximumFood	>=0

191	//SNMaximumWood	>=0
192	//SNMaximumStone	>=0
193	//SNMaximumGold	>=0
215	//SNAllowDiplomacyChangeOnAllyAttack	0,1
216	//SNMinimumAmountForTrading	>=0
217	//SNAllowDiplomacyChangeOnTribute	0,1
221	//SNHitsBeforeAllianceChange	>=0
178	//SNAttackDiplomacyImpact	0-100
218	//SNEasiestReactionPercentage	0-100
219	//SNEasierReactionPercentage	0-100
201	//SNTrackPlayerHistory	0,1
188	//SNAttackWinningPlayer	0,1
194	//SNCoopShareInformation	0,1
195	//SNAttackWinningPlayerFactor	>=0
196	//SNCoopShareAttacking	0,1
197	//SNCoopShareAttackingInterval	>=0
198	//SNPercentageExploreExterminators	0-100
186	//SNCoopDemandTributeInterval	>=0
187	//SNCoopDemandTributeMaximum	>=0

0-7[(i)] - 0=No Priority, 1=Highest, 7=Lowest

Item Id[(ii)] – Look through the .ai files in Folders Data &Data 2. All units can be found. First number is unit or technology ID.

[(iii)] - Unknown.